A Two-Phase Deep Learning Framework for Detecting Deepfake and AI-Generated Videos

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***Abstract*—The rapid advancement of generative artificial intelligence has significantly increased the creation of highly realistic synthetic videos, creating new challenges for digital trust, media verification, and misinformation prevention. While conventional deepfake detection systems primarily focus on distinguishing manipulated videos from authentic recordings, many existing approaches do not explicitly address fully AI-generated videos produced by modern generative models. This limitation reduces their practical applicability in current real-world environments where multiple forms of synthetic media coexist.**

This paper presents a two-phase video authenticity assessment framework designed to identify different categories of suspicious video content through a sequential decision pipeline. In Phase 1, uploaded videos are analyzed to determine whether they are AI-generated or camera-captured authentic recordings. This stage employs spatial-frequency forensic analysis using residual noise patterns and Fast Fourier Transform (FFT) representations, followed by classification using a ResNet-18 backbone. In Phase 2, videos classified as camera-captured content are further examined for facial manipulation artifacts to determine whether they are authentic or deepfake videos. This phase uses face extraction, multi-dataset training, and a weighted soft-voting ensemble of XceptionNet and EfficientNet-B2 models.

Experimental evaluation demonstrates strong performance across both phases. Phase 1 achieved 98.23% frame-level accuracy and 99.50% video-level accuracy, while the Phase 2 ensemble achieved 96.67% test accuracy. Results indicate that combining forensic AI-generated video detection with cross-dataset deepfake analysis provides a practical and scalable solution for modern video authenticity verification.

***Index Terms*—Deepfake Detection, AI-Generated Video Detection, Video Forensics, Computer Vision, Ensemble Learning, Synthetic Media Authentication**

I. Introduction

The authenticity of digital video content has become a major concern due to rapid progress in artificial intelligence driven media generation. Modern generative systems are capable of producing realistic videos containing synthetic humans, fabricated scenes, and altered visual content. At the same time, deepfake methods continue to evolve, enabling face swapping, facial reenactment, identity replacement, and lip synchronization manipulations that are increasingly difficult to detect visually.

These developments create serious risks in journalism, social media, cybersecurity, digital forensics, and personal privacy. Fabricated videos may be used for misinformation, impersonation, fraud, reputation damage, or manipulation of public opinion. Therefore, reliable automated systems for verifying video authenticity are becoming essential.

Most existing research focuses mainly on deepfake detection, where the objective is to classify a video as real or manipulated. However, this view is becoming insufficient in the generative AI era. A video may not only be a manipulated version of a real recording, but may also be fully AI-generated without any original camera-captured source. Hence, modern authenticity analysis must consider both deepfake manipulation and fully generated video content.

Another limitation of existing approaches is their dependence on single benchmark datasets. A model trained on one dataset may learn dataset-specific compression artifacts, lighting conditions, or manipulation traces, causing reduced generalization on unseen sources. To address this, the proposed work combines multiple important deepfake datasets and uses random subsamples to improve robustness across diverse manipulation styles.

This paper proposes a two-phase deep learning framework for video authenticity assessment. Phase 1 detects whether a video is AI-generated or camera-captured using residual and FFT-based forensic features. Phase 2 is applied only to camera-captured videos and determines whether the video is authentic or deepfake manipulated using face-based deep learning models and weighted ensemble prediction.

The main contributions of this work are as follows:

- A two-phase hierarchical framework for modern video authenticity verification.
- A Phase 1 AI-generated video detection method using residual and FFT-based forensic representations.

- A Phase 2 deepfake detection method trained using multiple benchmark datasets for improved generalization.
- A weighted soft-voting ensemble of XceptionNet and EfficientNet-B2 for robust deepfake classification.
- A practical Flask-based upload-and-detect interface for video-level prediction.

The rest of this paper is organized as follows. Section II presents related work. Section III explains the proposed methodology. Section IV describes the experimental setup. Section V discusses the results. Section VI presents limitations, and Section VII concludes the paper.

II. Related Work

Deepfake detection has received significant attention in recent years, especially with the release of benchmark datasets such as FaceForensics++, DFDC, and Celeb-DF. Early approaches focused on visible inconsistencies such as abnormal eye blinking, facial boundary artifacts, blending errors, and unnatural head pose. Later methods used convolutional neural networks to learn manipulation traces directly from video frames and facial crops.

XceptionNet has been widely used as a strong baseline for face forgery detection because depthwise separable convolutions can effectively capture subtle spatial artifacts introduced during synthesis and blending. EfficientNet variants have also been adopted because they provide a strong balance between accuracy and computational efficiency.

Some studies have explored temporal modeling using recurrent neural networks, optical flow, 3D CNNs, and transformer-based architectures. These methods can capture motion inconsistencies across frames, but they often require higher computational resources and longer training time. Due to limited GPU availability, this work focuses on efficient frame-level processing followed by video-level aggregation.

In parallel, AI-generated media detection has investigated synthetic content through frequency-domain artifacts, residual noise inconsistencies, and generator fingerprints. Residual filtering and Fourier analysis are useful because generative models often leave hidden traces in texture statistics and frequency spectra that may not be clearly visible in RGB space.

Despite these advances, many existing methods address only one form of synthetic media. They either detect manipulated deepfake videos or fully generated content separately. The proposed framework combines both objectives into a sequential pipeline, making it more suitable for real-world video authenticity verification.

III. Proposed Methodology

A. Overall System Architecture

The proposed system follows a two-phase decision pipeline. First, the uploaded video is processed by Phase 1 to determine whether it is AI-generated or camera-captured. If the video is classified as AI-generated, the system directly outputs the AI-generated label. If the video is classified as camera-captured, it is passed to Phase 2, where the system determines whether the video is authentic or deepfake manipulated.

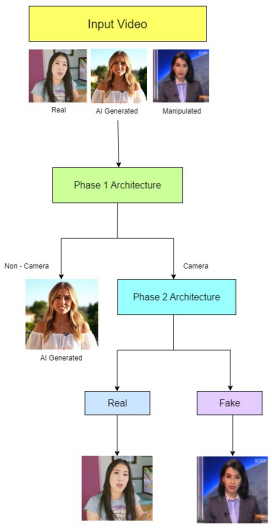

Fig. 1. Overall two-phase video authenticity assessment pipeline.

This hierarchical structure separates generation-level synthesis from manipulation-level forgery detection. The final output belongs to one of three classes: AI-generated video, authentic real video, or deepfake video.

B. Phase 1: AI-Generated vs Camera-Captured Video Detection

In Phase 1, each input video is divided into 15 evenly spaced frames. This sampling strategy captures representative content from the video while reducing computational cost. Each extracted frame is processed to generate two forensic representations: a residual image and an FFT image.

The residual image is generated by suppressing dominant visual content and highlighting fine-grained local differences. A Gaussian blurred version of the frame is subtracted from the original frame, producing a high-pass residual representation. This helps expose subtle noise patterns and local inconsistencies left by AI generation.

The FFT representation converts the frame into the frequency domain. The magnitude spectrum is computed after applying the Fourier transform and logarithmic scaling. This representation helps detect unnatural frequency patterns and repeated artifacts that may appear in AI-generated videos.

Fig. 2. Residual and FFT feature extraction used in Phase 1.

The processed representations are used as input to a ResNet-18 based classifier. ResNet-18 was selected because it is lightweight, efficient, and suitable for the available computational resources. The model predicts whether each frame belongs to the AI-generated or camera-captured class.

The video-level prediction is obtained by aggregating frame-level predictions using majority voting. For a video containing k sampled frames, the final label is computed as:

$$\hat{Y}_{video} = \text{mode}(y_1, y_2, y_3, ..., y_k) \tag{1}$$

where $y_1, y_2, ..., y_k$ are the predicted labels for the extracted frames.

C. Phase 2: Deepfake Detection

If Phase 1 classifies the video as camera-captured, the video is passed to Phase 2 for deepfake detection. Similar to Phase 1, 15 evenly spaced frames are extracted from each video. Since most deepfake manipulations occur in facial regions, MTCNN is used to detect and crop faces from the sampled frames.

The cropped facial images are used to train and evaluate two deep learning models: XceptionNet and EfficientNet-B2. XceptionNet is effective in learning facial manipulation artifacts such as blending errors, texture inconsistencies, and boundary distortions. EfficientNet-B2 is selected because it provides strong feature extraction capability while maintaining computational efficiency.

To improve prediction robustness, the outputs of both models are combined using weighted soft voting. The final ensemble probability is computed as:

$$P_{ensemble} = 0.55P_{Xception} + 0.45P_{EfficientNet} \tag{2}$$

The class with the highest ensemble probability is selected as the final Phase 2 prediction.

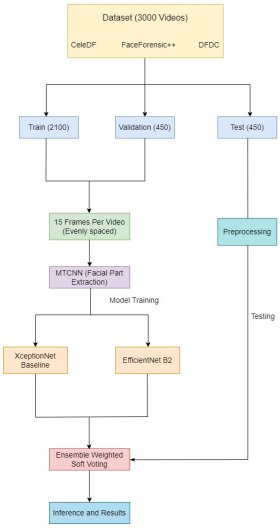

Fig. 3. Phase 2 deepfake detection using face extraction and weighted ensemble learning.

D. Deployment Interface

A Flask-based interface was developed to make the system practical for user-level testing. The user uploads a video through the interface, and the backend automatically performs frame extraction, feature generation, model inference, and final prediction generation.

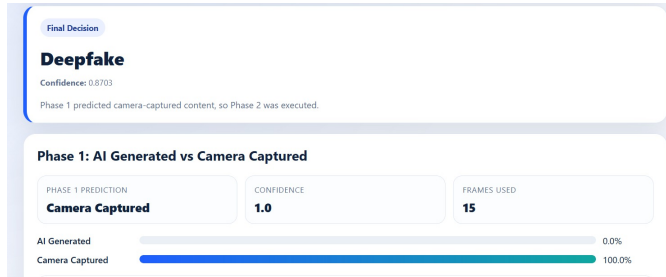

Fig. 4. User interface for uploading videos and viewing detection results.

IV. Experimental Setup

A. Phase 1 Dataset

The Phase 1 dataset contains camera-captured videos and AI-generated videos. Camera-captured samples were collected from real video sources, while AI-generated samples were created using modern generative video tools. The dataset was divided into training, validation, and testing sets as shown in Table I.

TABLE I
Phase 1 Dataset Split

Split	Number of Videos
Training	1400
Validation	200
Testing	200

Each video was represented using 15 evenly spaced frames. Therefore, the Phase 1 test set consisted of 3000 test frames from 200 test videos.

B. Phase 2 Dataset

The Phase 2 dataset was constructed by combining samples from multiple benchmark datasets: FaceForensics++, DFDC, and Celeb-DF. This multi-source strategy was used to reduce dataset-specific bias and improve generalization across different deepfake generation methods.

TABLE II
Phase 2 Dataset Split

Split	Number of Videos
Training	2100
Validation	450
Testing	450

The fake class includes multiple manipulation styles, while the real class contains authentic videos. Faces were extracted using MTCNN before model training.

C. Training Configuration

The models were trained using transfer learning. ResNet-18 was used in Phase 1, while XceptionNet and EfficientNet-B2 were used in Phase 2. Cross-entropy loss was used for binary classification. Adam optimizer with a learning rate of 1×10^{-4} was used during training. Validation accuracy and loss were monitored to select the best-performing model checkpoint.

Due to limited GPU resources, the system was designed with efficient frame sampling, lightweight preprocessing, and model selection strategies that reduce unnecessary computation.

D. Evaluation Metrics

The system was evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Phase 1 was evaluated at both frame level and video level. Phase 2 was evaluated using model-wise comparison between XceptionNet, EfficientNet-B2, and the ensemble model.

Frame-level accuracy is computed as:

$$\text{Accuracy}_{frame} = \frac{N_{correct_frames}}{N_{frames}} \times 100 \tag{3}$$

Video-level accuracy is computed as:

$$\text{Accuracy}_{video} = \frac{N_{correct_videos}}{N_{videos}} \times 100 \tag{4}$$

V. Results and Discussion

A. Phase 1 Results

The Phase 1 classifier achieved strong performance in distinguishing AI-generated videos from camera-captured videos. At the frame level, the model achieved a test accuracy of 98.23% with a test loss of 0.0631. Both AI-generated and camera-captured classes achieved balanced precision, recall, and F1-score values of 0.98.

TABLE III
Phase 1 Frame-Level Classification Report

Class	Precision	Recall	F1-score	Support
AI	0.98	0.98	0.98	1500
Camera	0.98	0.98	0.98	1500
Accuracy		98.23%		

At the video level, frame predictions were combined using majority voting. This improved reliability and reduced the effect of occasional frame-level errors. The model achieved 99.50% video-level accuracy.

TABLE IV
Phase 1 Video-Level Classification Report

Class	Precision	Recall	F1-score	Support
AI	1.00	0.99	0.99	100
Camera	0.99	1.00	1.00	100
Accuracy		99.50%		

These results show that residual and FFT representations are effective for detecting AI-generated video artifacts.

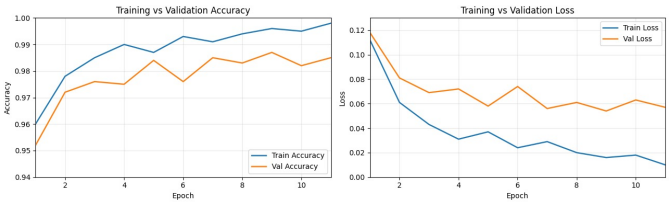

Fig. 5. Training and validation performance for Phase 1.

B. Phase 2 Results

Phase 2 was evaluated using XceptionNet, EfficientNet-B2, and their weighted ensemble. XceptionNet achieved 95.56% accuracy, while EfficientNet-B2 achieved 94.00% accuracy. The weighted ensemble achieved the highest accuracy of 96.67%.

TABLE V
Phase 2 Model Performance Comparison

Model	Test Accuracy
EfficientNet-B2	94.00%
XceptionNet	95.56%
Ensemble	96.67%

The detailed ensemble classification report is shown in Table VI.

TABLE VI
Phase 2 Ensemble Classification Report

Class	Precision	Recall	F1-score	Support
Fake	0.9605	0.9733	0.9669	225
Real	0.9730	0.9600	0.9664	225
Accuracy		96.67%		

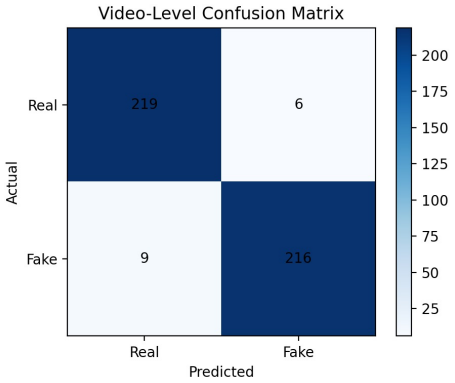

Fig. 6. Confusion matrix of the Phase 2 ensemble model.

C. Dataset-Wise Accuracy Analysis

The ensemble model performed strongly across different datasets. FaceForensics++ achieved the highest dataset-wise accuracy, while DFDC was comparatively more challenging due to greater variation in lighting, compression, pose, and manipulation quality.

TABLE VII
DATASET-WISE TEST ACCURACY COMPARISON

Model	Celeb-DF	DFDC	FaceForensics++
XceptionNet	96.25%	89.23%	100.00%
EfficientNet-B2	93.13%	87.69%	100.00%
Ensemble	96.88%	92.31%	100.00%

D. Discussion

The results demonstrate that the proposed two-phase framework is effective for modern video authenticity verification. Phase 1 successfully detects AI-generated videos using forensic residual and frequency-domain cues. Phase 2 improves deepfake detection through multi-dataset training and ensemble learning.

The ensemble model outperformed both individual models, indicating that XceptionNet and EfficientNet-B2 learn complementary facial manipulation features. XceptionNet performed better as a standalone model, but EfficientNet-B2 contributed additional robustness when combined through weighted soft voting.

The major development challenges were AI video data generation, dataset collection, and limited GPU resources. Despite these limitations, the proposed framework achieved strong results in both AI-generated video detection and deepfake detection.

VI. LIMITATIONS

This work has several limitations. First, the system mainly relies on frame-level analysis and does not explicitly model long-term temporal inconsistencies using LSTM, 3D CNN, or transformer-based temporal modules. Second, although multiple datasets were combined in Phase 2, unseen future deepfake generation methods may still require additional training data. Third, the ensemble model increases inference cost compared with a single-model approach. Finally, heavily compressed or low-resolution videos may reduce the visibility of forensic artifacts.

VII. CONCLUSION

This paper presented a two-phase deep learning framework for detecting AI-generated and deepfake videos. The proposed system first identifies whether a video is AI-generated or camera-captured using residual and FFT-based forensic features. If the video is camera-captured, it is further analyzed using a face-based deepfake detection pipeline.

Experimental results demonstrate that the proposed approach is effective for both stages of video authenticity analysis. Phase 1 achieved 98.23% frame-level accuracy and 99.50% video-level accuracy, while the Phase 2 ensemble achieved 96.67% test accuracy. These results show that combining forensic feature extraction, multi-dataset learning, and weighted ensemble classification provides a strong foundation for practical video authenticity verification.

Future work will focus on incorporating temporal modeling, expanding AI-generated video sources, improving robustness against compression, and adapting the system to emerging generative video models.

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